

# Spatial Resolution Tier Boundary Validation and Monadic Stock Transitions in Planetary Ecological Simulations (Sprint 24)

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Sprint 24 Technical Report

## Abstract

In global-scale ecological and biogeochemical simulations, maintaining spatial indexing integrity is a fundamental prerequisite for enforcing physical conservation laws. This paper summarizes the architecture and implementation of Sprint 24 within the WEB OF LIFE simulation engine. We introduce formal boundary check functions for Uber’s H3 hierarchical hexagonal spatial index system across its discrete resolution tiers ( $r \in [0, 15]$ ). By integrating these checks into monadic spatial state wrappers (**SpatialMonad**), we guarantee the rigorous enforcement of the First Law of Thermodynamics (exact mass conservation across hierarchical cell refinements) and the Second Law of Thermodynamics (proper solar energy flux density scaling).

## 1 Introduction and Thermodynamic Context

The WEB OF LIFE simulation models Earth as a closed thermodynamic system with respect to matter, while admitting open solar energy fluxes. Spatial domains are discretized over planetary surfaces using Uber’s H3 hierarchical hexagonal grid system.

Within this paradigm, spatial resolution tier bounds ( $r \in [0, 15]$ ) act as structural invariants. Unbounded or fractional resolutions would corrupt cell area calculations ( $A_r$ ), leading to unphysical singularities in solar radiation flux density and violating mass conservation during monad stock transitions.

## 2 Mathematical Framework & Delta Equations

### 2.1 Resolution Domain and Monad Structure

Let the valid set of resolution tiers  $\mathcal{R}$  be defined as:

$$\mathcal{R} = \{r \in \mathbb{Z} \mid 0 \leq r \leq 15\}$$

A spatial monad  $\mathcal{S}$  encapsulating ecological stock vector  $[B, E]$  (biomass  $B$ , energy  $E$ ) at H3 index  $H_3$  and resolution  $r$  is formally expressed as:

$$\mathcal{S} = (H_3, r, [B, E]) \quad \text{where } r \in \mathcal{R}$$

## 2.2 Conservation of Matter During Refinement

When a spatial monad undergoes hierarchical refinement from resolution  $r$  to  $r'$  ( $r' > r$ ), total biomass  $B$  is strictly conserved across child-parent cell allocations:

$$\Delta B_{\text{system}} = B_{\text{parent}} - \sum_{i=1}^k B_{\text{child},i} = 0$$

## 2.3 Solar Energy Flux Boundary Scaling

Energy flux density  $I$  received per cell area  $A_r$  at resolution  $r$  scales inversely with hexagonal cell area  $A_r$ :

$$E_{\text{solar}} = \Phi_{\odot} \cdot A_r \cdot \Delta t$$

Invalid resolutions ( $r \notin \mathcal{R}$ ) would result in undefined area scalings ( $A_r \rightarrow \text{NaN}$ ), violating energy conservation.

## 3 Implementation Architecture

The core validation logic is implemented in `src/spatial/h3_grid.ts` and integrated into `src/monads/spatial_monad.ts`:

```
validateResolutionTier(resolution: number): resolution is H3Resolution
```

This function ensures that only integers within  $[0, 15]$  propagate through spatial monad refinements, throwing an explicit `[SpatialError]` upon boundary violation.

## 4 Conclusion

Sprint 24 establishes robust spatial safety invariants for the WEB OF LIFE simulation engine. Future sprints will leverage these validated resolution tiers to execute high-resolution trophic energy transfers across planetary boundaries.

**Repository Reference:** <https://github.com/pascalranoroarijaona/WebOfLife>